



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 11 – 2010 YEARLY EXAMINATIONS

MATHEMATICS

ASSESSMENT TASK 4 WEIGHTING 40%

Monday 23rd August 2010
8.30 a.m. – 11.00 a.m.

OUTCOMES REFERRED TO: P2, P3, P4, P5, P6, P7

General Instructions

- Reading time – 5 minutes.
- Working time – $2\frac{1}{2}$ hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question.
- **Begin** each question on a **new page**.
- Write your **name**, **class** and your **teacher's name** on the front of each answer sheet of paper.

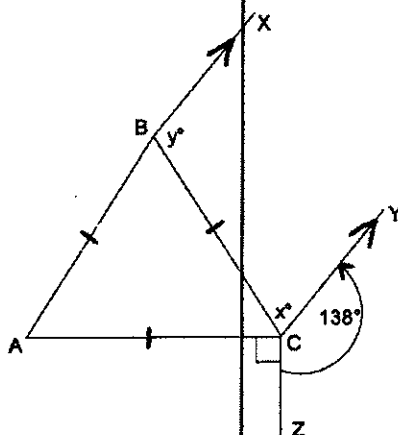
Total marks – 100

- Attempt Questions 1-10.
- All questions are of equal value.
- Mark values are shown at the side of each question part.

Question 1. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

- (a) Evaluate $\frac{(2.1)^3}{\sqrt{9.5-2.8}}$ correct to:
- (i) 2 decimal places. 1
- (ii) 2 significant figures. 1
- (b) An importer buys 1800 toy dolls for \$6240 and sells them at a profit of 50%. Calculate the selling price of each doll. 1
- (c) Factorise completely $12a^2 + 6a^3$ 1
- (d) Simplify $2\sqrt{20} - \sqrt{45}$ 2
- (e) Expand and simplify $(3y-2)(y+1)$. 2
- (f) Find the values of x and y . 2



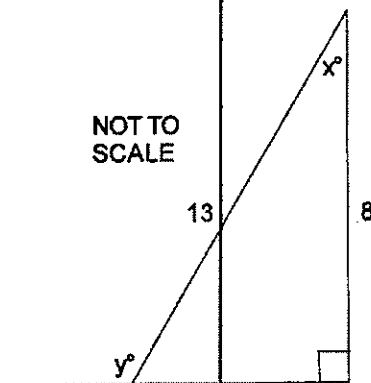
Question 2. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

(a) If $f(x) = 2x + |x|$, find the value of $f(-2)$. 1

(b) Express $0.4\dot{2}$ as a fraction in its simplest form. 2

(c) Find the values of x and y to the nearest degree. 2



(d) Express $\frac{(3k-4)}{2} - \frac{(k-2)}{5}$ as a single fraction in its simplest form. 2

(e) Solve the pair of simultaneous equations.

$$5m - 3n = 21$$

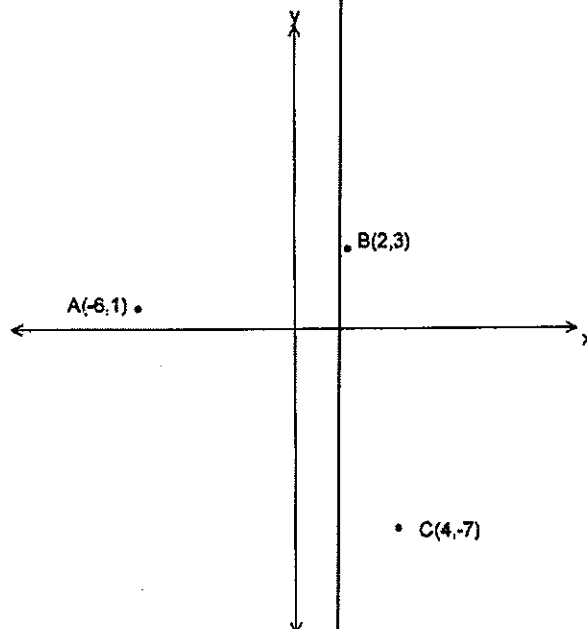
$$3m + n = 7$$

3

Question 3. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

In the diagram, the points $A(-6, 1)$, $B(2, 3)$, and $C(4, -7)$ are the vertices of a triangle.



- | | | |
|-----|---|---|
| (a) | Find the exact length of AB in simplest surd form. | 2 |
| (b) | Find the gradient of AB . | 1 |
| (c) | Show that the equation of the line AB is $x - 4y + 10 = 0$. | 2 |
| (d) | Find the equation of the line passing through the point C that is parallel to the y -axis. | 1 |
| (e) | Find the perpendicular distance from C to the line AB .
Leave your answer in exact form. | 2 |
| (f) | Hence, or otherwise find the area of triangle ABC . | 1 |
| (g) | Find the point D such that $ABCD$ is a parallelogram. | 1 |

Question 4. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

(a) Simplify $\tan\theta \operatorname{cosec}\theta$.

1

(b) State the domain of the function $y = \frac{1}{x-2}$.

1

(c) State the range of the function $y = 4 - x^2$.

1

(d) Rationalise the denominator of $\frac{6}{\sqrt{3}+1}$.

2

(e) Solve $4 - 3x < 6(x - 1)$

2

(f) Indicate on a diagram using shading, the region for which the following inequalities hold simultaneously:

$y \leq x^2 - 1$ and $x^2 + y^2 < 4$.

3

Question 5. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) Differentiate from first principles $f(x) = 3x^2 - 2x + 5$. 3

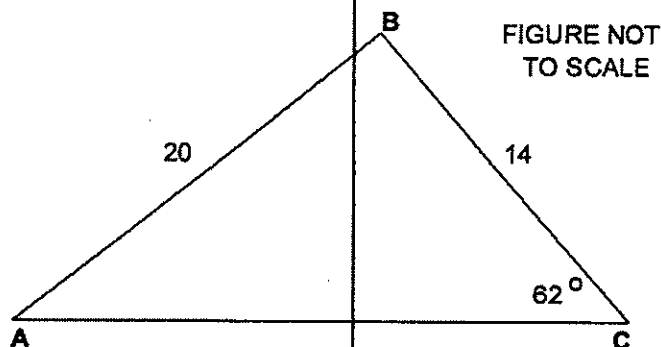
(b) Differentiate with respect to x :

(i) $2x^5 - 4x^2 + 3$ 1

(ii) $(4x+5)^4$ 1

(iii) $\frac{3x^2+1}{x}$ 1

(c)



(i) Find the size of $\angle BAC$ to the nearest degree. 2

(ii) Find the area of $\triangle ABC$. 2

Question 6. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

(a) If α and β are the roots of the equation $2x^2 - 3x - 6 = 0$, find the value of:

(i) $2\alpha + 2\beta$. 1

(ii) $\alpha^2\beta^2$. 1

(iii) $\alpha^2 + \beta^2$. 2

(b) Find the exact value of $\tan 240^\circ \times \sin 135^\circ$ 2

(c) Simplify $2 - 2\sin^2 \theta$. 1

(d) Solve the equation: $2\cos \theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. 2

(e) Sketch the graph for $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$. 1

Question 7. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

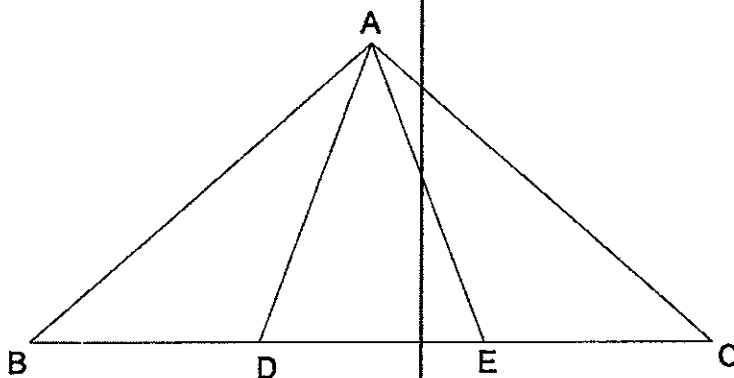
(a) Factorise $5a^2 + a - 18$. 2

(b) Find the values of x for which $|2x - 1| \geq 5$. 2

(c) (i) Sketch the graphs of $y = 3 - x$ and $y = |x| - 1$. 2

(ii) Write down the coordinates of all, if any, points of intersection of $y = 3 - x$ and $y = |x| - 1$. 1

(d) Given $AB = AC$ and $\angle BAD = \angle CAE$.



(i) Prove that $\triangle BAD$ and $\triangle CAE$ are congruent. 2

(ii) Hence or otherwise, prove that $\triangle DAE$ is isosceles. 1

Question 8. (10 marks) Start each question on a new page.

Write your name and your teacher's name at the top of each page.

(a) Show whether the function $f(x) = x^3 - x$ is odd, even or neither. 2

(b) Find the values of k for which the quadratic equation $x^2 - (k+2)x + 4 = 0$ has no real roots. 2

(c) Simplify $\frac{3x^2 + 7x + 2}{x^2 + 4x + 4} \div \frac{9x^2 - 1}{3x^2 - 3xy - x + y}$ 3

(d) Find the values of A , B and C for the identity $3x^2 - 16x + 17 \equiv A(x-2)^2 + B(x-2) + C - 5$ 3

Question 9. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

- (a) Two trees on a street have heights of 9 m and 3 m respectively. At midday the shorter tree casts a shadow of 5 m. How far apart are the trees? **2**

(b) Solve $\frac{x+4}{7} = \frac{x+5}{4}$ **2**

(c) Solve the pair of simultaneous equations.
 $2x - y = 5$
 $y = x^2 - 4x$ **3**

(d) Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + 2x - 8}$. **3**

Question 10. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) Sketch the function $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 3x & \text{if } 0 \leq x < 2 \\ 6 & \text{if } x \geq 2 \end{cases}$. 3

(b) Find the equation of the tangent to the curve $y = \frac{x+4}{x+1}$ at the point where $x = 2$. 3

(c) A helicopter travels from base station S on a bearing of 235° for 58 km to Outpost A. It then travels 135 km on a bearing of 340° to outpost B.

(i) How far is outpost B from base station S correct to 1 decimal place? 2

(ii) Find the bearing of outpost B from base station S to the nearest degree. 2

End of paper

YR 11 - T&S MATHEMATICS (2U) AUGUST 2010

Q1. a) i) 3.577836439
 3.58 (1)

ii) 3.6 (1)

b) $\frac{6240 \times 1.5}{1800} = 5.20$ (1)

c) $6a^2(2+a)$ (1)

d) $4\sqrt{5} - 3\sqrt{5}$ (M1)
 $= \sqrt{5}$ (1)

e) $3y^2 + 3y - 2y - 2$ (M1)
 $= 3y^2 + y - 2$ (1)

f) $x = 72^\circ$ (1)
 $y = 108^\circ$ (1)

$$\begin{aligned} \text{Q2. a) } f(-2) &= 2(-2) + |-2| \\ &= -4 + 2 \\ &= -2 \end{aligned} \quad (1)$$

$$\begin{aligned} \text{b) } 10x &= 4.222\dots \\ x &= 0.422\dots \\ \hline 9x &= 3.8 \quad (M1) \\ x &= \frac{38}{90} = \frac{19}{45} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{c) } \cos x &= \frac{8}{13} \\ x &= \cos^{-1}\left(\frac{8}{13}\right) \\ &= 52^\circ \quad (1) \\ \therefore y &= 142^\circ \quad (1) \end{aligned}$$

$$\begin{aligned} \text{d) } \frac{5(3k-4)}{10} - \frac{2(k-2)}{10} \\ &= \frac{15k-20-2k+4}{10} \quad (M1) \\ &= \frac{13k-16}{10} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{e) } 5m - 3n &= 21 \\ 9m + 3n &= 21 \quad (M1) \\ \hline 14m &= 42 \\ m &= 3 \quad (1) \\ n &= 7 - 3(3) \\ &= 7 - 9 \\ n &= -2 \quad (1) \end{aligned}$$

$$\begin{aligned} \text{Q3. a)} \quad AB &= \sqrt{(2+6)^2 + (3-1)^2} \quad (M1) \\ &= \sqrt{68} \\ &= 2\sqrt{17} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{b)} \quad m_{AB} &= \frac{3-1}{2+6} \\ &= \frac{1}{4} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{c)} \quad y-3 &= \frac{1}{4}(x-2) \quad (M1) \\ 4y-12 &= x-2 \\ x-4y+10 &= 0 \quad (1) \end{aligned}$$

$$\text{d)} \quad x = 4 \quad (1)$$

$$\begin{aligned} \text{e)} \quad d &= \frac{|(1)(4) - 4(-7) + 10|}{\sqrt{1+16}} \quad (M1) \\ &= \frac{42}{\sqrt{17}} \quad \text{OR} \quad \frac{42\sqrt{17}}{17} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{f)} \quad A &= \frac{1}{2} \cdot 2\sqrt{17} \times \frac{42}{\sqrt{17}} \\ &= 42 \text{ units}^2 \quad (1) \end{aligned}$$

$$\text{g)} \quad D(-4, -9) \quad (1)$$

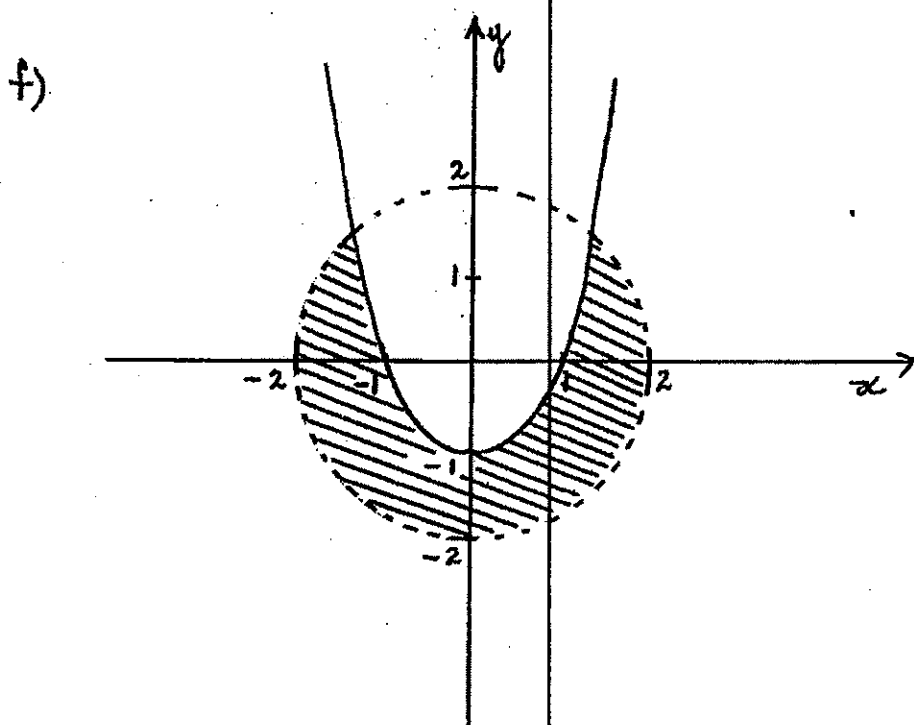
Q4. a) $\frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\sin \theta}$
 $= \sec \theta$ ①

b) Domain: all real x , $x \neq 2$ ①

c) Range: all real x , $y \leq 4$ ①

d) $\frac{6}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)}$
 $= \frac{6(\sqrt{3}-1)}{2}$ (M1)
 $= 3(\sqrt{3}-1)$ ① OR $3\sqrt{3}-3$

e) $4 - 3x < 6x - 6$
 $-9x < -10$
 $x > \frac{10}{9}$



Q5.

a)

$$\begin{aligned}
 & \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{3(x^2 + 2xh + h^2) - 2(x+h) + 5 - (3x^2 - 2x + 5)}{h} \quad (M1) \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6xh + \cancel{3h^2} - \cancel{2x} - 2h + 5 - \cancel{3x^2} + \cancel{2x} - \cancel{5}}{h} \quad (M1) \\
 &= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 2)}{h} \\
 &= 6x - 2 \quad (1)
 \end{aligned}$$

b) i) $10x^4 - 8x \quad (1)$

ii) $4(4x+5)^3(4)$
 $= 16(4x+5)^3 \quad (1)$

iii) $\frac{d}{dx} [3x + x^{-1}]$
 $= 3 - x^{-2}$
 $= 3 - \frac{1}{x^2} \quad (1)$

c) i) $\frac{\sin A}{14} = \frac{\sin 62^\circ}{20}$
 $A = \sin^{-1} \left[\frac{14 \sin 62^\circ}{20} \right] \quad (M1)$
 $A = 38^\circ \quad (1)$

ii) $\therefore \angle B = 180 - (62 + 38)$
 $= 80^\circ \quad (1)$

$$\begin{aligned}
 A &= \frac{1}{2} (20)(14)(\sin 80^\circ) \\
 &= 137.8730854 \\
 &\doteq 138 \text{ units}^2
 \end{aligned}$$

Q6. a) $\alpha + \beta = \frac{3}{2}$

i) $2(\alpha + \beta)$
 $= 2\left(\frac{3}{2}\right)$
 $= 3$ (1)

$\alpha\beta = -3$

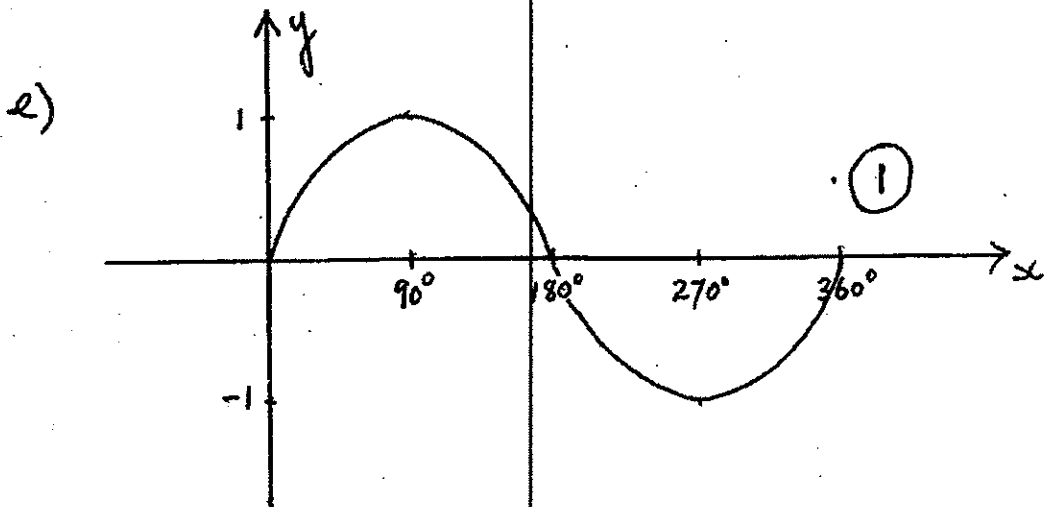
ii) $(\alpha\beta)^2$
 $= (-3)^2$
 $= 9$ (1)

iii) $\alpha^2 + \beta^2$
 $= (\alpha + \beta)^2 - 2\alpha\beta$ (M1)
 $= \left(\frac{3}{2}\right)^2 - 2(-3)$
 $= 8\frac{1}{4}$ (1)

b) $\tan 240^\circ \times \sin 135^\circ$
 $= \sqrt{3} \times \frac{1}{\sqrt{2}}$ (M1)
 $= \frac{\sqrt{3}}{\sqrt{2}}$ (1) or $\frac{\sqrt{6}}{2}$

c) $2(1 - \sin^2 \theta)$
 $= 2\cos^2 \theta$ (1)

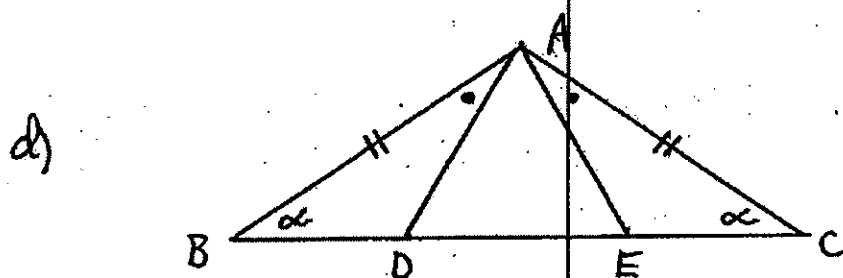
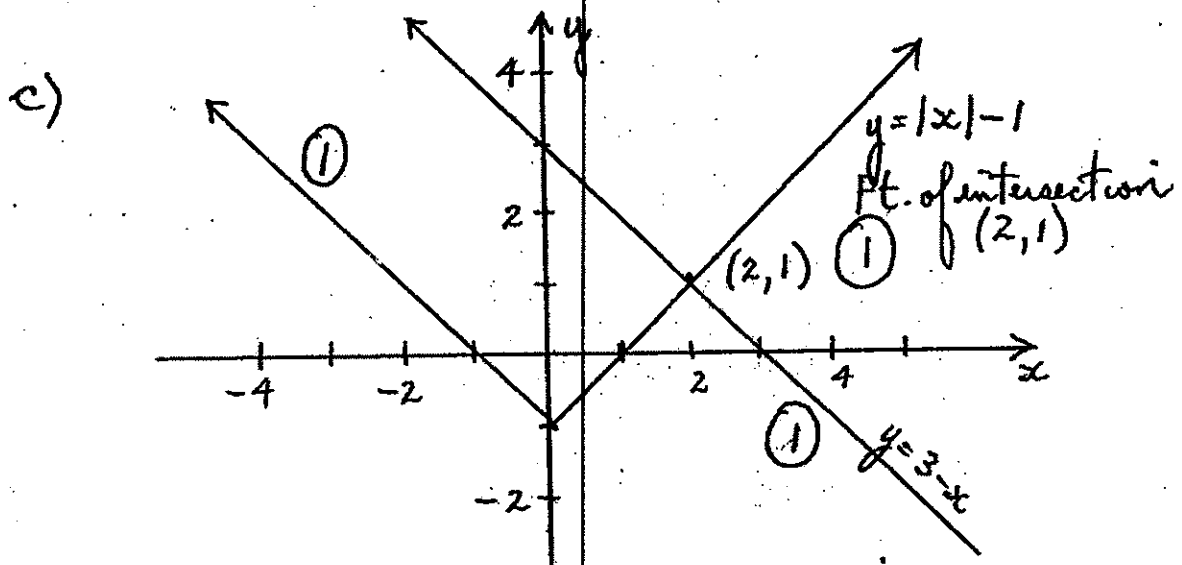
d) $\cos \theta = \frac{1}{2}$
 $\theta = 60^\circ, 300^\circ$ (1) (1)



Q7. a) $5a^2 + a - 18$
 $= (5a - 9)(a + 2)$
 (1) (1)

b) $2x - 1 \geq 5$
 $2x \geq 6$
 $x \geq 3$ (1)

$2x - 1 \leq -5$
 $2x \leq -4$
 $x \leq -2$ (1)



i) $AB = AC$ Given
 $\angle B = \angle C$ Base $\angle$'s isos. Δ . =. (MI)
 $\angle BAD = \angle CAE$ Given
 $\Delta BAD \equiv \Delta CAE$ A.A.S (1)

ii) $AD = AE$ Cor. sides $\equiv \Delta$'s =. (1)
 $\therefore \Delta DAE$ is isosceles

Q8.

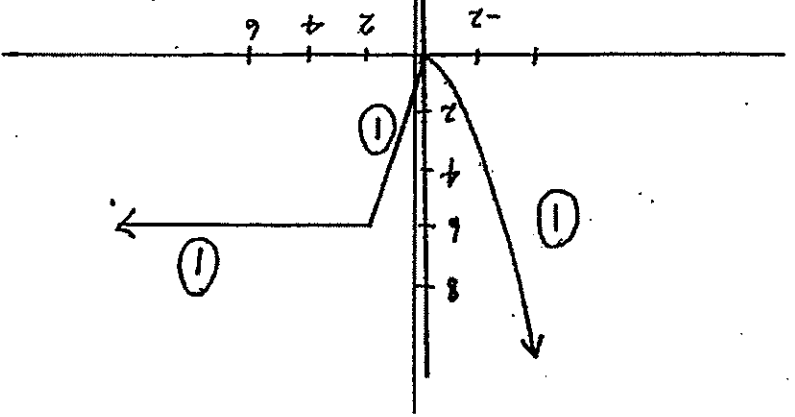
a) $f(x) = x^3 - x$
 $f(-x) = -x^3 + x$
 $= -(x^3 - x)$ (M1)
 $\therefore$ since $f(-x) = -f(x)$ function is odd (1)

b) $b^2 - 4ac < 0$
 $(k+2)^2 - 4(1)(4) < 0$
 $k^2 + 4k + 4 - 16 < 0$ (M1)
 $k^2 + 4k - 12 < 0$
 $(k+6)(k-2) < 0$
 $\therefore -6 < k < 2$ (1)

c) $\frac{(3x+1)(x+2)}{(x+2)(x+2)} \times \frac{(3x-1)(x-y)}{(3x+1)(3x-1)}$ (M1)
 $= \frac{x-y}{x+2}$ (1)

d) $3x^2 - 16x + 17 \equiv A(x^2 - 4x + 4) + Bx - 2B + C - 5$
 $\equiv Ax^2 - 4Ax + 4A + Bx - 2B + C - 5$
 $\equiv Ax^2 + (B - 4A)x + 4A - 2B + C - 5$
 $\therefore A = 3$ (1)
 $B - 4A = -16$
 $B - 12 = -16$
 $B = -4$ (1)
 $4A - 2B + C - 5 = 17$
 $12 + 8 + C - 5 = 17$
 $C = 2$ (1)

Q10. a)



$$f_y = \frac{(x+1)(1) - (x+4)(1)}{(x+1)^2 - (x-4)^2}$$

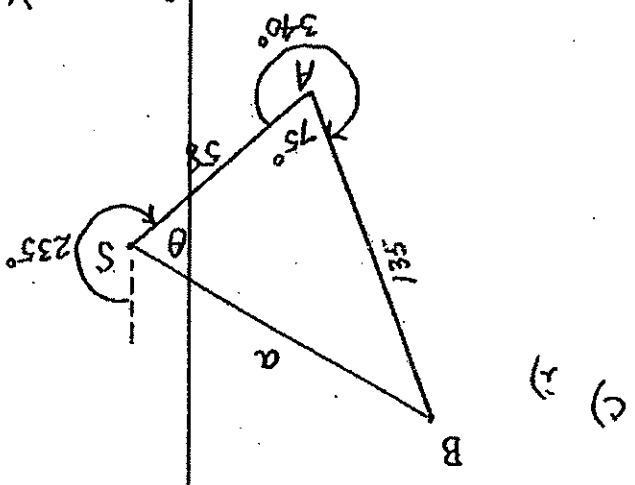
$$= \frac{(x+1)^2}{(x+1)^2 - (x-4)^2} \quad (M1)$$

$$= \frac{-3}{-3} = 1 \quad (M1)$$

$$4-2 = -2 \quad (x-2)$$

$$3y-6 = -2(x+2)$$

$$x+3y-8=0 \quad (1)$$



$$a^2 = 58^2 + 135^2 - 2(58)(135)(\cos 75^\circ) \quad (M1)$$

$$a = 132.4 \text{ km} \quad (1)$$

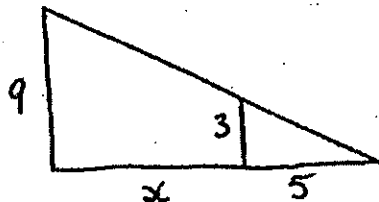
$$ii) \frac{\sin \theta}{135} = \frac{\sin 75^\circ}{132.4}$$

$$\theta = \sin^{-1} \left[\frac{135 \sin 75^\circ}{132.4} \right]$$

$$\theta = 80^\circ \quad (1)$$

Bearing is $235 + 80$ (1)

Q9. a)



$$\frac{3}{9} = \frac{5}{x+5} \quad (M1)$$

$$3x + 15 = 45$$

$$3x = 30$$

$$x = 10 \quad (1)$$

b)

$$4(x+4) = 7(x+5) \quad (M1)$$

$$4x + 16 = 7x + 35$$

$$-19 = 3x$$

$$x = -\frac{19}{3} \quad (1)$$

c)

$$y = 2x - 5$$

$$y = x^2 - 4x$$

$$\therefore x^2 - 4x = 2x - 5 \quad (M1)$$

$$x^2 - 6x + 5 = 0$$

$$(x-5)(x-1) = 0$$

$$x = 5 \quad (1)$$

$$y = 5 \quad (1)$$

$$x = 1 \quad (1)$$

$$y = -3 \quad (1)$$

d)

$$\lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+4)} \quad (M1)$$

$$= \lim_{x \rightarrow 2} \frac{x^2+2x+4}{x+4} \quad (M1)$$

$$= \frac{12}{6}$$

$$= 2 \quad (1)$$